JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 95, NO. B5, PAGES 6737-6750, MAY 10, 1990

PALEOMAGNETISM OF EOCENE VOLCANIC ROCKS,TALKEETNA MOUNTAINS, ALASKA

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Abstract. Previous paleomagnetic studies of Eocene rocks deposited in the Talkeetna Mountains on the northern edge of the combined Peninsular, Wrangellia, and Alexander terranes (the southern Alaska superterrane (SAS)) yield a paleolatitude of 76°N, while Late Cretaceous rocks on the southern edge of the SAS record a 32°N paleolatitude. At face value, these drastically different paleolatitudes imply that Wrangellia moved 44° to the north in only 20 m.y., requiring a northward component of velocity of about 24 cm/yr. Alternatively, the discrepancy might be explained by postulating an unrecognized tectonic boundary separating the two localities, allowing different emplacement ages. We have tested the unrecognized tectonic boundary hypothesis (allowing post-Eocene emplacement of the SAS) by sampling Eocene volcanic rocks deposited near the southern edge of the SAS in the Talkeetna Mountains. Our data set comprises measurements from 97 oriented cores collected from 26 lava flows and two tuffs distributed over a distance of 50 km. A primary remanence is indicated by positive fold tests and penecontemporaneous intrusion tests. Age control is provided by nine new K-Ar whole rock age determinations ranging from 38.8 to 53.6 Ma. The mean paleomagnetic pole calculated for this study is not significantly different from the pole derived from the Eocene flows to the north and gives a paleolatitude of 78°, thus ruling out the hypothesis of an unrecognized suture accommodating major post-Eocene displacement. This result has prompted a reevaluation of the data which suggest rapid movement, including the precision of the age control, the adequacy of averaging of secular variation, and the possibility of sedimentary inclination error. The bulk of the evidence suggests northward displacement from lower latitudes; however, the error limits are such that the dramatic velocities previously postulated are not required.

Introduction

It is generally accepted that much of southern Alaska comprises displaced terranes which originated in low paleolatitudes [e.g., Jones et al., 1977; Hillhouse, 1977]. However, the age(s) of final emplacement of these terranes continues to be a subject of considerable debate. Resolution of the accretion age problem has been confounded by unrecognized remagnetizations, imprecise age control, and contradictory geological mapping relationships. This has prompted several workers to revise the data sets and use only those paleomagnetic studies which pass field stability tests for tectonic analysis [e.g., Coe et al., 1985; Panuska and Stone, 1985]. Unfortunately, even with culled data sets, apparent contradictions remain.

Paleomagnetic studies suggest that interior Alaska, inboard of the Denali fault system (Figure 1), was only slightly displaced from the south in latest Cretaceous time [Globerman, 1985; Coe et al., 1985]. By Paleocene time, interior Alaska was at its present latitude with respect to North America [Hillhouse and Grommé, 1982; Coe et al., 1985]. Rocks from south of the Denali fault in the Peninsular terrane, part of the southern Alaska superterrane (SAS), yielded paleolatitudes suggesting only slight displacement with respect to North America, during latest Cretaceous to earliest Paleocene time [Coe et al., 1985; Thrupp and Coe, 1986]. (The southern Alaska superterrane (SAS), a name proposed by Panuska [1984, 1985], is the superterrane consisting of the Peninsular, Wrangellia, and Alexander terranes. This name is preferred over the name Talkeetna superterrane [Csejtey et al., 1982] because this latter name does not include the Alexander terrane, an integral part of the superterrane.) Additionally, Eocene rocks which overlap the northern edge of the SAS show no paleomagnetically observable displacement [Hillhouse et al., 1985]. This led Thrupp and Coe [1986] to suggest that the boundary between displaced and largely in-place terranes occurs south of the SAS (Figure 1). However, latest Cretaceous rocks from the Wrangell Mountains (SAS) yielded an anomalously low paleolatitude [Panuska, 1985], suggesting large-scale northward displacement This finding is apparently at odds with studies which show little or no Tertiary movement.

One possible explanation of this discrepancy is that the sedimentary rocks sampled in the Wrangell Mountains study may contain an inclination error, a misrecording of the magnetic field inclination due to sedimentary processes, leading to a systematically low estimate of the paleolatitude as suggested by Coe et al. [1985]. This possibility is presently under investigation. Nevertheless, the paleomagnetic data from the outboard terranes [Plumley et al., 1983; Hillhouse and Grommé, 1977; Bol and Coe, 1987], which appear to be geologically tied to the SAS by latest Cretaceous to earliest Tertiary time, tend to support the low paleolatitude estimate from the Wrangell Mountains. Although low paleolatitudes

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Paper number 89JB03404.
0148-0227/90/89JB-03404$05.00

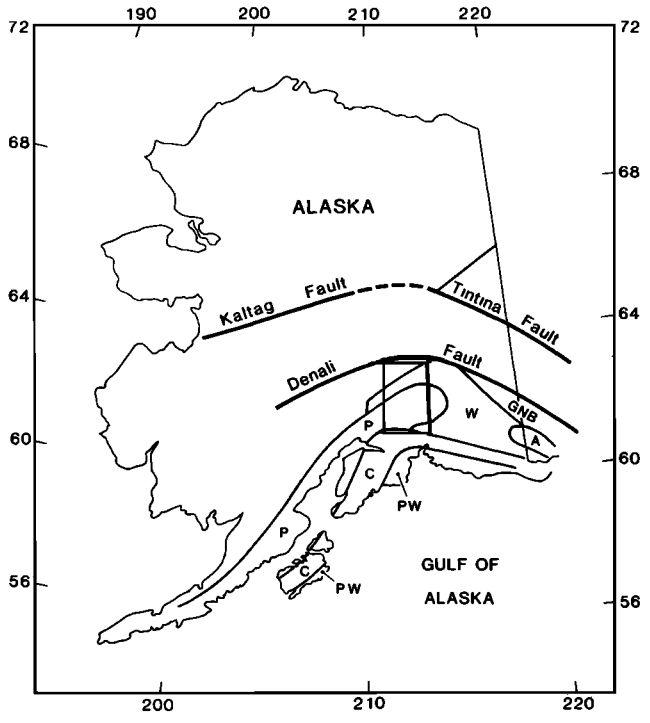


Fig. 1. Generalized terrane location map. P, Peninsular terrane; W, Wrangellia terrane; A, Alexander terrane; GNB, Gravina-Nutzotin Belt overlap terrane; C, Chugach terrane; PW, Prince William terrane. Rectangle indicates location of Figure 2.

obtained for the Wrangells and outboard terranes show rough agreement, these results conflict with the Peninsula terrane data from Thrupp and Coe [1986].

Alternatively, the paleolatitude discrepancies might be explained as an artifact of insufficient age control. Some of the ages of magnetization were constrained by K-Ar dates with high analytical uncertainty [Thrupp and Coe, 1986], correlation with other dated exposures [Plumley et al., 1983], uncertain correlations [Hillhouse and Grommé, 1977], and circumstantial evidence from sparse magnetostratigraphic data [Panuska, 1984, 1985]. Although it is unlikely that any of these ages are in error by tens of millions of years, the lack of age precision may have significant consequences. If the face value paleolatitudes and ages for the Late Cretaceous and Eocene Wrangellia data [Panuska, 1985; Hillhouse et al., 1985] are correct, northward components of motion of approximately 24 cm/yr are implied. This very high rate of motion corresponds to nearly 1000 km (approximately 10°) of movement in only 4 m.y. Since ages of the paleomagnetic sites are seldom known to be better than 4 m.y., large paleolatitude discrepancies between localities can be expected within the error limits.

The present study was undertaken in part to investigate the lack of agreement between some of these data sets and in part to expand the areal extent of the data base. In addition, data sets once thought to pass stability tests satisfactorily are now known to be significantly contaminated by regional overprinting [Haeussler and Coe, 1987; Coe et al., 1985; Panuska and Decker, 1985]. Therefore it is important to conduct additional paleomagnetic studies in order to verify the existing work.

Several localities of unnamed Eocene lava flows were sampled for paleomagnetic analysis primarily to compare the results with a paleomagnetic study of age equivalent rocks in the northern Talkeetna Mountains [Hillhouse et al., 1985]. Although the study by Hillhouse et al. [1985] persuasively argues for a stable magnetization, representing terranes within and to the north of the SAS, additional work is warranted because of the following:

1. With the exception of one locality, Hillhouse et al. [1985] sampled rocks occurring north of the SAS. Field mapping does suggest stratigraphic continuity of the volcanic unit into the SAS. Nevertheless, given the conflicting evidence, a test of geologic correlations and paleomagnetic data is in order.

2. The study of Hillhouse et al. [1985] had formation wide age control but relatively little age control at the sample localities.

3. Hillhouse et al. [1985] measured a steep magnetic inclination (82°) corresponding to a 76° paleolatitude. Paleolatitude estimates are very sensitive to inclination measurement at steep values. Incomplete sampling of secular variation might give rise to substantial changes in paleolatitude estimates [e.g., Panuska and Macicak, 1986].

4. Because numerous authors have identified significant counter-clockwise rotation of much of southern and southwestern Alaska [e.g., Coe et al., 1985], and because rotations are very difficult to identify at high latitudes, increasing the temporal and geographic data base is important to tectonic models.

Sampling Plan

Geological mapping in the southeastern Talkeetna Mountains was carried out by Grantz [1960a, b, c, 1961], Magoon et al. [1976], and Detterman et al. [1976]. Based on this mapping and on logistic considerations, we sampled five localities over a distance of approximately 50 km (Figure 2). At these localities, reliable paleomagnetic data were acquired from 26 lava flows and two tuffs.

Paleomagnetic sampling was guided by several criteria. First, there had to be local indication of ancient horizontal. In general, this attitude control was afforded by flow contacts; however, in some cases sedimentary interbeds or stratified pyroclastic rocks were available. Exposures had to be extensive enough to allow recognition of flows, dikes, and sills, so that samples could be grouped by individual cooling units and so that complications due to intrusives could be avoided. The sampled flows are between 4 and 10 m thick. Flows were usually recognized by massive basalt or andesite grading upward into flow breccia, often with vesicular texture. A special effort was made to sample the massive, nonvesicular flow interiors as far from flow contacts as possible, because samples from flow breccia clasts could give spurious directions.

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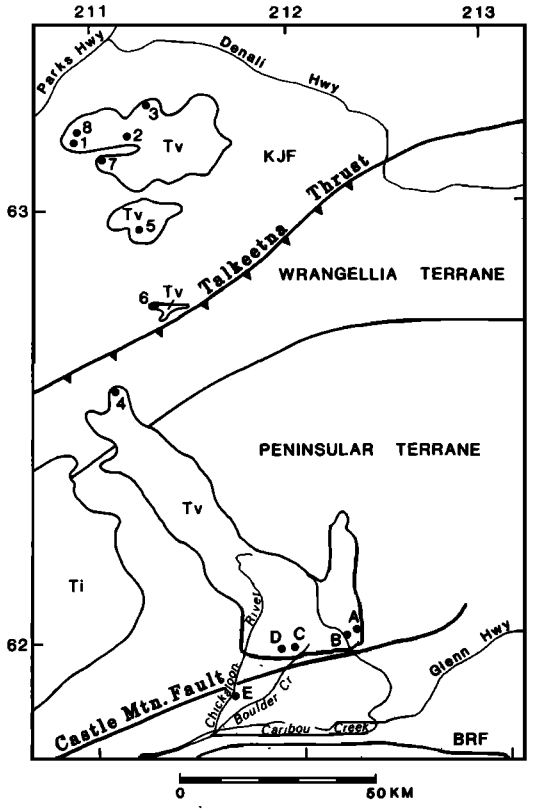


Fig. 2. Sample locality map for Eocene volcanic rocks in the Talkeetna Mountains. Localities 1-8 are from Hillhouse et al. [1985]. Localities A-E refer to areas sampled for this study. KJF is Jurassic-Cretaceous flysch, Ti is Early Tertiary intrusive, and Tv is Eocene volcanic units.

In addition to the flows and tuffs, a few dikes (thought to be penecontemporaneous with the flows [see Silberman and Grantz, 1984]) were sampled in order to apply intrusion tests for remagnetization. The stratigraphic directions (corrected for tectonic tilt) for dikes were determined by applying the same structural correction used for the nearest flow. Since there is no way to infer the original attitude of the dikes reliably, these were excluded from the calculation of mean paleomagnetic directions and are included in the data table only for the purpose of comparison. Typical sample density used in this study is three to five cores per cooling unit. One exception is flow E-7, which has only one core due to unsafe field sampling conditions. Cores were oriented by magnetic compass, checked by sighting to known landmarks and, whenever possible, by sun compass readings. One specimen cut from each core was used in the paleomagnetic analysis, except where replicate specimens where needed to resolve problems encountered in analysis or interpretation of results.

K-Ar Dating

The geochronology of the Tertiary volcanic rocks in the southern Talkeetna Mountains has been reported by Silberman and Grantz [1984]. These ages range between 56.2 and 43.6 Ma, although one questionable age of 60.1 Ma was obtained [Silberman and Grantz, 1984]. In order to provide tighter constraints on the age of magnetization, samples for whole rock 40_K-40_{AR} dating were taken from selected paleomagnetic collection sites.

Samples with varying amounts of alteration of potassium-bearing phases or with interstitial glass are reported as minimum ages and are indicated by asterisks in Table 1. Two samples with abundant alteration (38BP85, SE136, Table 1) were treated with 5% HF, followed by 14% HNO_3 and ultrasonic cleaning in distilled water in an attempt to remove most of the alteration products. Each sample was dated before and after acid treatment. Both acid-treated splits gave significantly older ages than their untreated counterparts, indicating that acid treatment removed a potassium-bearing phase that was not retentive to radiogenic argon (Table 1). The acid-treated splits gave ages at or near the age of the volcanic sequence determined by other dated samples which meet petrographic criteria for reliable whole-rock K-Ar dating. For these reasons, the ages of the two untreated splits of 38BP85 and SE136 were not included in our age analysis of the volcanic sequence.

Three K-Ar samples were analyzed from locality A (Figure 2). One sample collected near the base of the section (corresponding to flow A-2) has an age of 44.1 ± 1.3 (2σ) Ma. Two samples collected from flow A-8, at the top of the 150-m-thick A section, yield minimum ages of 46.9 ± 2.8 Ma and 43.5 ± 2.6 Ma. These ages are in general accord with the nearest age (8 km away) reported by Silberman and Grantz [1984] of 48.4 ± 2.4 (1σ) Ma. The B locality is located only 2 km away from locality A and was sampled at about the same stratigraphic level. The age equivalence of the two localities is supported by minimum ages of 47.1 ± 2.8 Ma (flow B-3) and 48.4 ± 2.9 Ma (flow B-2).

The Cretaceous age determinations for flow B-1 can not be accurate as well-established stratigraphic relationships preclude ages older than Paleocene. One sample of flow B-1 yielded a minimum age of 104.5 ± 6.3 Ma. A reanalysis of the same sample gave a similar age, 97.0 ± 5.8 Ma (minimum). These unrealistic ages prompted the analysis of a new sample from the same flow giving a minimum age of 73.6 ± 4.4 Ma. It is not presently known why these samples apparently contain excess radiogenic argon.

Localities C and D are located along upper Boulder Creek (Figure 2). A sample from the lower portion of the local volcanic section (flow C-3) gives a reliable age of 53.6 ± 3.2 Ma. Flow C-1 gives a minimum age of 41.4. ± 2.4 Ma. Locality D is located about 4 km from locality C and is about 300 m higher in the section. Flow D-2 yields a reliable age of 40.8 ± 2.4 Ma. A sample from D-3, immediately above flow D-2, has a minimum age of 39.1 ± 2.4 Ma. The ages for localities C and D are similar to ages for nearby rocks (50.5 ± 1.5 Ma, 45.5 ± 1.8 Ma, 51.4 ± 1.5 Ma) reported by Silberman and Grantz [1984].

Locality E, Puddingstone Hill, lies south of the Castle Mountain fault system. The Puddingstone Hill flows overlie the late Paleocene-early Eocene Wishbone Formation. Samples from flows E-1 and E-4 yield minimum ages of 39.6 ± 2.6 Ma and 38.8 ± 2.4 Ma, respectively. Dikes and sills from the general area have been dated at 37.5 ± 1.2 Ma to 43.5 ± 1.7 Ma [Silberman and Grantz, 1984].

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TABLE 1. Analytical Data for Whole Rock Basalt ^{40}K-^{40}AR Age Determinations

Flow	Field	Mean K_2O wt %	$^{40}Ar_{rad}$, %	$^{40}Ar_{rad}$, mol/g x 10^{-11}	Age, Ma ($\pm 2\sigma$)
			Locality A		
A-8	39BP85	0.380	54.2	2.41	$\geqq 43.5 \pm 2.6$*
A-8	38BP85	0.474	51.9	2.66	$\geqq 38.6 \pm 2.4$*
A-8	38BP85	0.494	56.0	3.38	$\geqq 46.9 \pm 2.8$* (acid treated)
A-2	35BP85	0.489	72.2	3.14	44.1 ± 2.6
			Locality B (Approximately Stratigraphically Equivalent to A)		
B-3	85BCP352 A	0.513	46.4	3.53	$\geqq 47.1 \pm 2.8$*
B-2	85BCP346 B	0.551	68.3	3.89	$\geqq 48.4 \pm 2.9$*
B-1	41BP85	0.552	14.1	8.54	$\geqq 104.5 \pm 6.3$*
B-1	41BP85	0.552	18.0	7.84	$\geqq 97.0 \pm 5.8$*
B-1	85BCP365 A	0.561	23.0	6.07	$\geqq 73.6 \pm 4.4$*
			Locality D		
D-3	SE136	0.185	3.2	0.635	$\geqq 23.7 \pm 3.8$*
D-3	SE136	0.158	42.7	0.900	$\geqq 39.1 \pm 2.4$* (acid treated)
D-2	SE132	0.335	66.3	1.99	40.8 ± 2.4
			Locality C (Stratigraphically Below D)		
C-3	SE120	0.436	22.5	3.41	53.6 ± 3.2
C-1	SE101	0.096	25.7	0.547	$\geqq 39.2 \pm 2.4$*
C-1	SE101	0.085	26.9	0.513	$\geqq 41.4 \pm 2.4$* (acid treated)
			Locality E (South of Castle Mountain Fault; Stratigraphic Relationship to Other Localities Unknown)		
E-4	SE159	0.239	30.1	1.35	$\geqq 38.8 \pm 2.4$*
E-1	SE150	0.090	25.4	0.518	$\geqq 39.6 \pm 2.6$*

rad, radiogenic; σ, standard deviation; $\lambda_\varepsilon + \lambda'_\varepsilon$, 0.581×10^{-10} yr^{-1}; $^{40}K/K$ total, 1.167×10^{-4} mol/mol. Flows are listed in probable stratigraphic order (youngest at top of table). ^{40}K-^{40}Ar dating was done at the Geophysical Institute-Alaska Divsion of Geological and Geophysical Surveys Cooperative Geochronology Laboratory in Fairbanks.
*Minimum age due to alteration.

The radiometric analyses available for the Tertiary volcanic rocks in the southern Talkeetna Mountains indicate ages ranging from early to middle (and possible late) Eocene. The weighted mean age for the volcanics, based on the dates reported here, may be taken as approximately 45 Ma. Although these ages span only 14 m.y., future tectonic investigations may require additional radiometric analyses to provide tighter age control. This may be especially important if models involving rapid motion are substantiated.

Paleomagnetic Analysis

Most magnetic measurements were made using an SCT cryogenic magnetometer (Z-Axis, Inc., California), although a few of the more strongly magnetized specimens were measured on a Schonstedt SM-1 spinner magnetometer (University of Alaska). In general, natural remanent magnetization (NRM) intensities were approximately 1 A/m (expressed as dipole moment per volume M [see Shive, 1986]), although a few

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NRMs ranged as high as 100 A/m. A suite of 11 pilot specimens was subjected to thermal cleaning, and 11 companion specimens from the same flows were cleaned by alternating field (AF) demagnetization. Most specimens responded equally well to either AF or thermal demagnetization. However, one flow, B-2, appeared to have two polarities present. It was thought that this might be due to two components of magnetization with overlapping coercivity spectra. Since the thermal demagnetization seemed to show more promise for separating the components and because unblocking temperature data could be more useful in other studies, the decision was made to thermally demagnetize the remaining specimens.

Knowledge of the magnetic behavior of the specimens gained from the pilot study was used to select the most appropriate demagnetization temperatures. Each specimen was demagnetized in 12-18 steps, with remanence measurements being made after each step. In order to insure proper characterization of the remanence directions, demagnetization intervals of 10°-40°C were used at the temperatures where intensity changes were most pronounced. The large majority of specimens demagnetized in a straightforward fashion. Representative demagnetization plots from two flows, each with thermal and AF cleaned specimen pairs, are shown in Figure 3. The vector diagrams show nearly straight-line decay of magnetization direction toward the origin, indicating a single-component magnetization. In a few specimens, secondary components were present; however, these components were generally removed by temperatures of 410°C or less (Figure 4).

Magnetic unblocking temperatures for flow A-3 specimens are relatively high (500-530°C) and coercivity spectra are relatively narrow (Figure 3). This suggests that low-Ti titanomagnetite is the carrier of remanence. In contrast, the remanence from the A locality tuff unblocks between 600°C and 620°C (well in excess of magnetite Curie temperature) and the coercivity spectrum is considerably broader than those of the flows (Figure 3). Such behavior is probably the result of remanence being carried by both magnetite and hematite.

Six of the 29 flows sampled were directionally

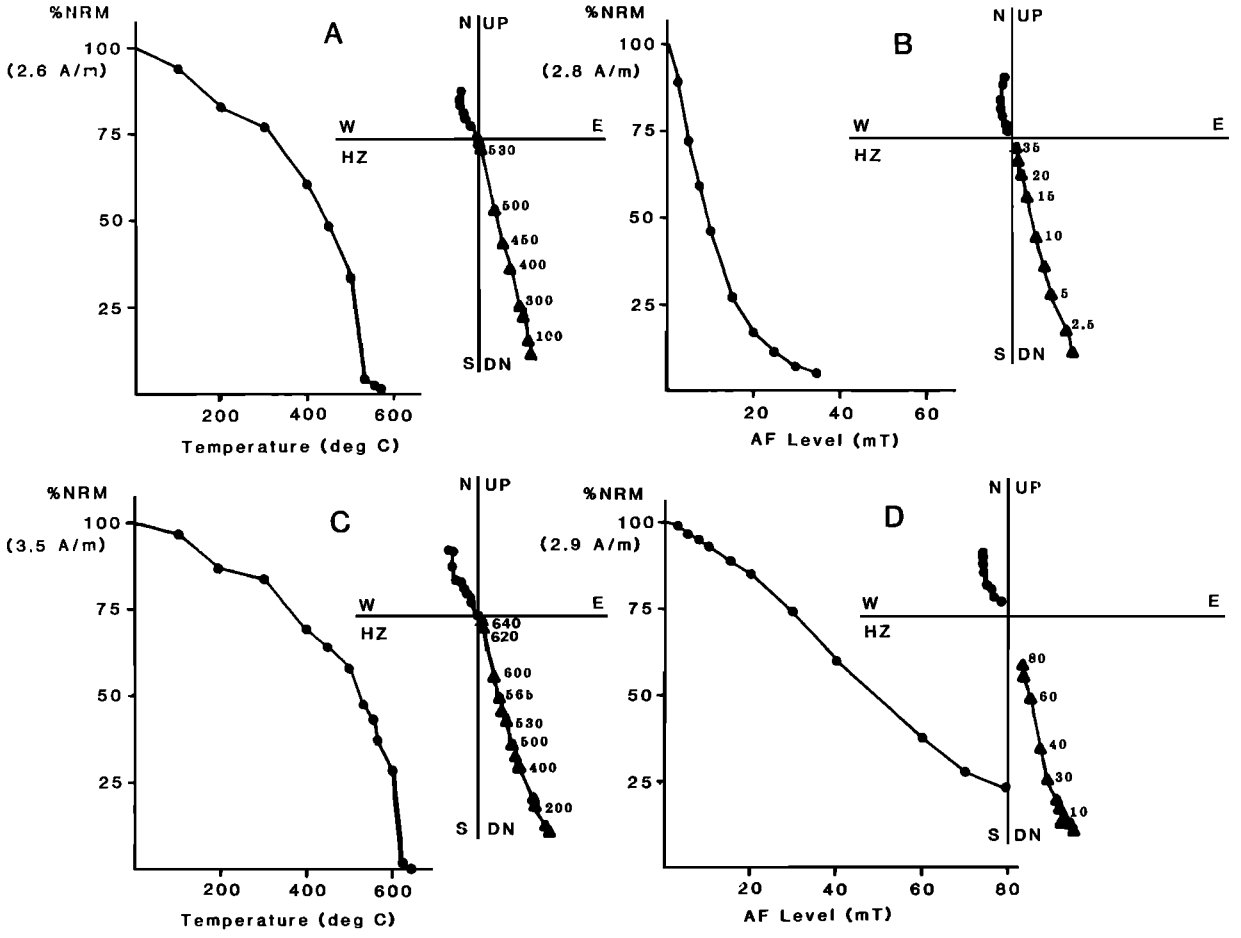


Fig. 3. Demagnetization diagrams and orthogonal vector component plots showing demagnetization behavior of four representative specimens. Figures 3a and 3b show thermal and AF cleaning results (respectively) for samples of flow A-3. Figures 3c and 3d show thermal and AF cleaning results (respectively) for tuff samples form locality A. Dots represent declination plotted in N, E, S, W plane. Triangles represent inclination plotted in vertical plane through the vector. (HZ, horizontal; DN, down).

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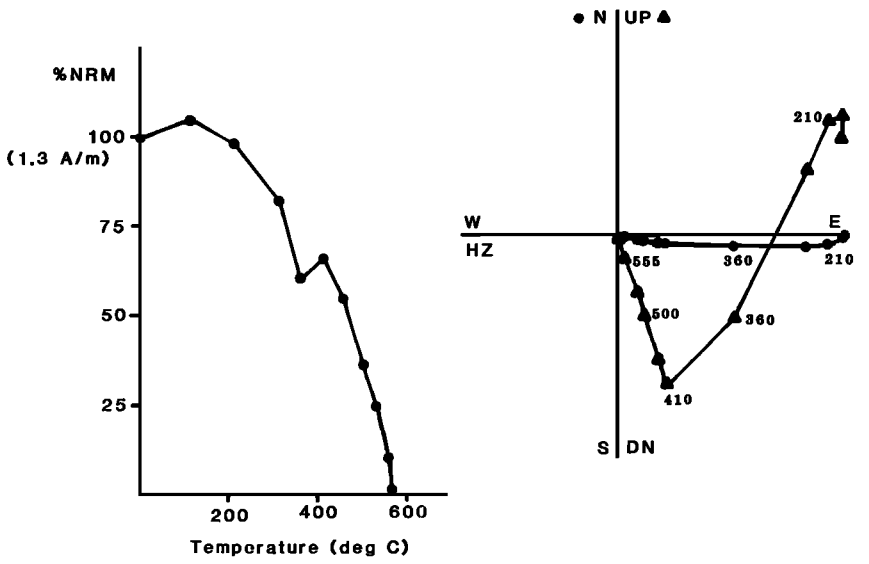


Fig. 4. Demagnetization diagram and orthogonal vector component plot (flow A-6) showing two components of magnetization. Secondary component of magnetization is removed by demagnetization at 410°C. See Figure 3 for conventions.

stable to temperatures of 670°C. Many of the specimens for these flows retained 20-60% of NRM intensity at temperatures exceeding the magnetite Curie point, indicating a significant contribution of hematite to the total remanence. The measured unblocking temperatures for the remainder of the flows varied between 500°C and 575°C, which is characteristic of Ti-poor titanomagnetite and magnetite.

In an attempt to characterize the magnetic mineralogy more completely, several magnetic separates were made for X ray analysis. X ray diffraction patterns were made using CuK_α radiation, at a 1°/min scan rate. Cell dimensions were calculated based on four or five measurements of the (311) titanomagnetite peak for each separate. The X ray data yielded cell dimensions between 8.407Å and 8.393Å, which corresponds to ulvospinel mole fractions of X = 0.13 to end-member magnetite [O'Reilly, 1984]. The Curie temperatures for titanomagnetites in this compositional range agree well with the observed unblocking temperature range of 500°-575°C. This suggests that low Ti-titanomagnetites are the predominant magnetic carriers.

Analysis of Paleomagnetic Data

The characteristic remanence directions derived from the progressive demagnetization data were averaged by cooling unit in order to determine the mean direction and statistical error limits of the various units (Table 2). These flow means were then used to evaluate the stability of the paleomagnetic directions, so as to guard against interpreting overprinted directions as primary directions. The stability tests applied to this data set consisted of a penecontemporaneous intrusion test and a fold test.

The intrusion test is a simple comparison of mean remanence directions obtained from flows and from dikes intruding the flows. If the magnetic directions of the dikes and their intruded flows are the same, then there is a distinct possibility that the intrusion of the dike caused a remagnetization of the flow or that both the dike and flow were remagnetized by some later event. If the remanence directions of the dike and flow are different, we conclude that no remagnetization of the flow or flow and dike has occurred since the time of dike intrusion. Thus, for a flow to be remagnetized and have a different signature than the dike, the resetting of remanence would have to occur before dike intrusion. The closer the age of dike intrusion is to the age of the flow, the less likely it is that the flow has been remagnetized due to the diminished time window available for remagnetization to occur. Therefore a penecontemporaneous intrusion test is a relatively strong test of stability.

Figure 5 shows the mean directions of two dike/flow sets at locality A. In both cases, the mean remanence directions of the dikes and their associated flows are significantly different, using the statistical technique of McFadden and Jones [1981]; thus they pass the intrusion test.

The intrusion test at locality B is more problematic. The dike gives a reversed polarity direction and both normal and reversed polarities are observed in the adjacent flow (Figure 6a), indicating that at least some of the specimens are overprinted. The flow (B-2) specimens that are cleaned by AF have characteristic directions which are much the same as their NRM directions. In contrast, the specimens cleaned by thermal demagnetization show a low temperature reversed magnetic component and normal components at higher temperatures. Unfortunately, remanence directions from most of these specimens fail to converge on a stable endpoint direction (Figure 6b). The change to normal polarity at high demagnetization temperatures suggests that the reversed direction is the overprinted direction.

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TABLE 2. Mean of Flows/Beds

Locality/ Flow	Unblocking Temperatures	N	GEO				STRAT				VGP			
			D	I	K	α_{95}	D	I	K	α_{95}	Longitude	Latitude	K	α_{95}
A-1	575°C	4	240	89	642	3.6	304	82	604	3.7	174	67	177	6.9
A-2	575	4	149	78	816	3.2	181	84	913	3.0	212	51	223	6.2
Dike	575+	4	3	86	199	6.5	325	79	182	6.8	164	74	52	12.9
A-3	530	3	176	80	1455	3.2	334	76	1313	3.4	145	78	573	5.2
Tuff	575	4	158	81	267	5.6	346	76	227	5.5	137	83	84	10.1
A-4	530	4	184	74	631	3.7	309	79	570	3.9	163	69	186	6.8
Dike	565+	3	352	85	1482	3.2	344	61	1304	3.4	66	68	956	4.0
A-5	565	4	129	74	149	7.5	131	82	146	7.6	230	50	38	15.1
A-6	500-555	3	116	71	200	8.7	89	78	191	8.9	254	55	61	15.9
A-7	500	3	126	76	233	8.1	320	83	222	8.3	183	71	62	15.8
A-8	500	3	156	82	21	27.9	305	88	20	28.0	208	63	63	53.8
		9/32	147	79	115	4.8	339	86	71.8	6.1	204	69	19.9	11.8
B-1	575	5	147	81	94	7.9	211	78	56	10.3	197	42	17	19.1
B-2	575+	3	148	69	87	13.3	180	77	42	19.3	213	38	13	35.9
Dike	565+	3	334	-81	316	6.9	56	-78	310	7.0	5	-45	97	12.6
B-3	565	5	254	85	45	11.6	269	71	45	11.6	159	49	19	18.1
B-4	565	4	24	81	399	4.6	313	76	406	4.6	148	69	122	84
B-5	555	5	106	85	308	5.2	269	81	279	5.5	178	57	81	10.2
		4/19	104	88	102.5	9.1	268	79	69.1	11.1	174	56	23.2	19.5
C-1	545	5	221	85	71	9.1	97	83	73	9.0	237	58	21	17.1
C-2	545	4	331	70	52	12.8	Ø	71	54	12.6	37	83	20	21.0
C-3	570	3	220	83	71	14.7	122	84	70	14.8	229	54	18	29.7
C-4	570	3	230	63	59	16.1	216	72	63	15.7	189	33	26	24.8
C-5	575*	3	246	66	488	5.6	240	76	550	5.3	179	42	188	9.0
		5/18	253	77	27.1	15.0	238	88	28.3	14.6	204	59	8.3	28.2
D-1	570	3	227	69	51	17.4	221	70	51	17.4	185	31	19	29.5
D-2	570+	4	158	53	1.7	-	155	51	1.7	-	226	9	1.4	-
Dike	545+	3	197	57	11	39.2	192	57	11.0	39.2	199	14	8.3	46.0
D-3‡														
D-4	570	3	189	75	402	6.2	178	75	468	5.7	212	35	156	9.9
Sill‡														
		2/6	211	73	79.3	28.4	203	74	72	29.8	198	34	25.1	52.2
E-1	570	3	329	59	96	12.7	358	71	89	13.1	39	84	32	22.4
E-2	575	3	329	60	258	7.7	0	72	246	7.9	31	85	89	13.1
E-3	575*	3	301	46	56	16.6	304	64	53	17.2	121	56	27	24.0
E-4	575*	3	310	52	165	9.6	303	77	184	9.1	155	65	54	16.9
E-5	575*	3	314	50	128	10.9	313	75	126	11.0	142	69	41	19.5
Tuff	575	3	318	72	28	23.8	129	83	27	24.3	229	51	8	47.7
E-6	575	3	312	53	46	18.4	325	70	47	18.1	116	70	19	29.5
E-7	575	1	307	58	-	-	319	75	-	-	139	71	-	-
		8/22	314	57	69.9	6.7	327	76	50.7	7.9	147	75	16.2	14.2
Mean of flow tuff means		28/97	283	82	15.1	7.3	302	84	38.0	4.5	188	66	11.7	8.3

N, define; D, declination in degrees; I, inclination in degrees; K, Fisler precision parameter; α_{95}, circle of 95% confidence; GEO, in situ vector directions; STRAT, vectors corrected for structural tilt; VGP, virtual geomagnetic pole.

+ Rejected from calculation.

* Stable to 670°C.

‡ No stable end point.

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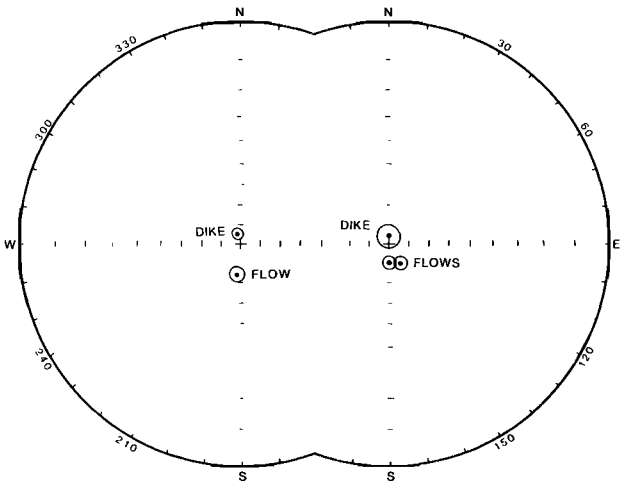


Fig. 5. Intrusion tests for dikes and flows at locality A. Dikes and the flows they intrude are magnetized in statistically different directions indicating that the dikes did not remagnetize the flows. Solid dots indicate positive inclination. Large circles are 95% confidence limits.

The normal polarity characteristic directions from flow B-2 form a somewhat elongate distribution (Figure 6a). This could be the result of an unresolved secondary component, although it could simply be a fortuitous alignment. Since the evidence for overprinting in some specimens is clear and since there is no evidence that all secondary components have been removed from the normal polarity characteristic directions, the mean direction for flow B-2 (reported in Table 2) is not included in subsequent mean calculations.

A more powerful test for primary magnetic remanence directions is the fold test. Figure 7 displays the mean of flow/tuff directions for each locality with the exception of locality D, which has only two usable flow directions and consequently a very large circle of 95% confidence. In the geographic (in situ) frame of reference, there is a substantial dispersion of mean directions, although there is some overlap of circles of confidence. The directions in the stratigraphic frame of reference, corrected for tilt, show a distinct clustering and a much greater overlap of circles of confidence than in the geographic frame. This strongly suggests a positive fold test, indicating that magnetization was acquired prior to tilting.

To determine the statistical significance of this fold test, the techniques outlined by McFadden and Jones [1981] were applied. These consist of comparing the mean directions of different limbs of a fold in both frames of reference. Although there is no well defined fold axis mapped between the localities we sampled, the mean directions for each locality may still be compared. Each locality is taken as approximating a homoclinally dipping section (see Table 3 for structural attitude data). By applying the McFadden and Jones [1981] statistical test on pairs of localities, the following can be concluded. Localities A + B, A + C and B + E pass the fold test indicating that these localities acquired their magnetization before tilting. The fold test between localities B and C is indeterminate because the difference in structural

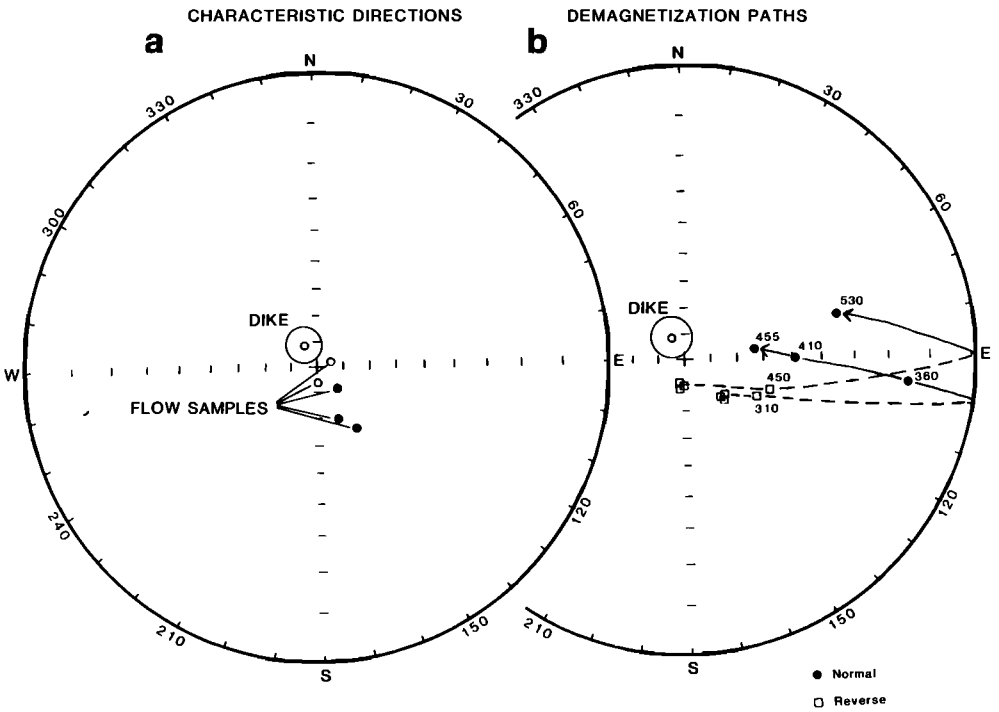


Fig. 6. Intrusion test for flow B-2. Negative inclination specimens (open symbols) are overprinted directions. See text for discussion. (a) Characteristic directions. (b) Demagnetization paths.

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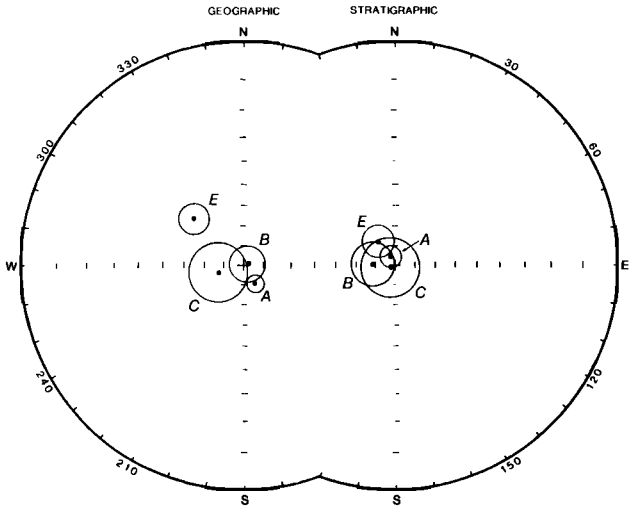


Fig. 7. Stereographic plot of localities used for fold test. Geographic frame of reference shows in situ directions. Stratigraphic frame of reference shows directions corrected for tilt. Marked convergence of locality mean vector directions in the stratigraphic frame of reference indicates that the magnetization predates folding (a positive fold test).

attitudes is not sufficient to produce a statistical difference between directions in either frame of reference. The fold test performed on localities A + E and C + E are positive at 99% confidence but indeterminate at 95% confidence. (It should be noted that the geographic directions are different regardless of the level of confidence). Since the McFadden and Jones test is designed to show a dissimilarity of directions at a desired level of confidence, the 95% level of confidence is a more stringent test of similarity than the 99% level of confidence. Due to the limited numbers of flows sampled at each locality and since secular variation at each locality may be incompletely averaged, the less stringent 99% confidence level is preferred. Accepting this argument leads to five positive fold test pairs and one indeterminate fold test pair and no negative fold test pairs. The fold tests and intrusion tests argue strongly for a primary remanence of all localities.

The mean stratigraphic direction calculated from the reliable flow and tuff data of all 5 localities is 302° declination, 84° inclination, $K = 38.0$, $\alpha_{95} = 4.5°$, and N = 28/97 (number of flows and tuffs/number of specimens). The mean paleomagnetic pole derived from these data is 188° E longitude, 66°N latitude, $K = 11.7$, $\alpha_{95} = 8.3°$. The circular standard deviation of the poles is 23.4°, which is a value consistent with secular variation models for high latitude sites. This suggests that the full range of secular variation has been sampled and that there is no bias in geomagnetic field observations due to temporally restricted sampling.

Before interpreting the tectonic significance of these data one additional reliability factor should be considered. Since lava flows require a topographic gradient in order to flow, it is possible that part of the structural correction applied to these data represents an initial dip. There are no obvious volcanic vent structures in the immediate vicinity of the sample localities. This suggests that the amount of primary dip represented in the various flow contacts is minimal. Circumstantial evidence for a lack of substantial primary dip at locality D comes from direct observation of the flow units, traceable in cliff exposures over several kilometers. These flows have no visible change in thickness and maintain constant attitude. To evaluate the possibility of a systematic bias of the data, we examined the distribution of structural attitudes. If one assumes that the magnitude of initial dip bias is the same for each flow but of

TABLE 3. Structural Attitudes of Flows

Flow	Azimuth	Dip
A-1	307	8
A-2	307	8
A/-3	343	23
tuff	342	24
A-4	342	24
A-5	306	8
A-6	326	10
A-7	311	21
A-8	329	10
B-1	247	10
B-3	275	14
B-4	275	14
B-5	275	14
C-1	76	10
C-2	76	10
C-3	76	10
C-4	76	10
C-5	76	10
D-1	98	3
D-4	98	3
E-1	115	18
E-2	115	18
E-3	115	18
E-4	135	25
E-5	135	25
Tuff	135	25
E-6	165	23
E-7	165	23

*Attitude convention is azimuth, bearing of downdip direction; dip, angle of flow with respect to horizontal.

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different direction, due to randomly distributed topographic slopes, and if the flows are sampled over a relatively large area, then there should be a tendency for the variable directions of the bias to offset each other. We averaged the poles to bedding for each flow or tuff used in the mean to see whether there are enough bedding attitudes in the data set to average out a possible initial dip bias. The mean pole to bedding is less than 2° from vertical, indicating that the number of dips in one direction is similar to the number of dips in the opposite direction. Thus, if there is a bias present in the data set due to initial dip of the flows, the effect of the bias should be randomized by the variations among the flow attitudes. While this is an admittedly crude means of evaluating the contribution of initial dip to the mean paleomagnetic direction, it does afford some degree of confidence that a given structural attitude has not been undersampled.

Discussion

The Eocene paleomagnetic pole position determined in this study is in excellent agreement with the pole position determined by Hillhouse et al. [1985] (Figure 8). This agreement and the large number of cooling units sampled in these studies (54 units total) suggests that secular variation has been adequately averaged out. The coincidence of pole positions confirms the geologic ties between the SAS and the terrane north of the Talkeetna thrust. This finding is significant given the problems in establishing agreement between geologic ties across various terrane boundaries and paleomagnetic measurements of displacement. Rather than plunging into tectonic scenarios and speculation, we will summarize the displacement

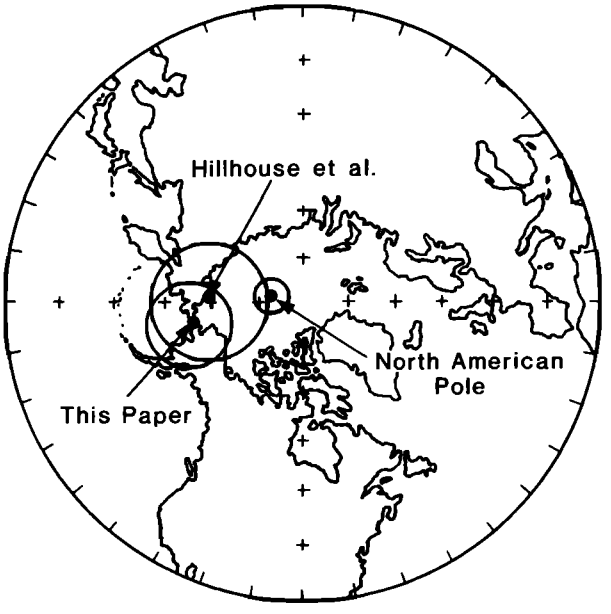


Fig. 8. Plot of paleomagnetic pole positions with circles of 95% confidence for Eocene volcanic rocks in the Talkeetna Mountains. Early/middle Eocene North America reference pole is from Diehl et al. [1983].

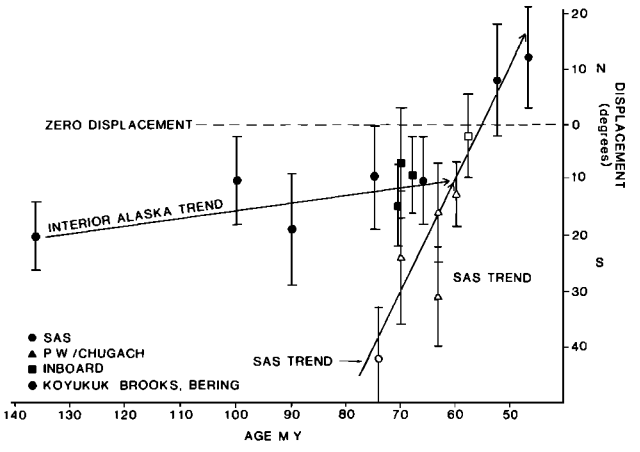


Fig. 9. Displacement versus time plot for all Alaska terranes. The data suggest two generalized displacement trajectories (relative to the North American Craton). The interior trend indicates little or no displacement during the Cretaceous and minor northward displacement in the earliest Tertiary for terranes inboard (north) of the Denali fault. The SAS trend suggests rapid northward displacement during the Late Cretaceous and early Tertiary. The SAS trend is not well-established due to uncertainties (open symbols) in age control, adequate averaging of secular variation or inclination error.

data for the localities within the SAS (Figure 9 and Table 4) and point out some of the more critical uncertainties.

The Resurrection Peninsula paleomagnetic data of Hillhouse and Grommé [1977] has recently been confirmed by Bol and Coe [1987]. The uncertainty in mean displacement is derived from the poor age control and therefore whether the 60 Ma or 70 Ma North American reference pole should be used. The different choices of reference poles would indicate 13° and 24° of displacement, respectively.

The Ghost Rocks Formation data may suffer from uncertainties in both age control and the averaging of secular variation. The Kiliuda Bay sample locality [Plumley et al., 1983; Coe et al., 1985] has both normal and reversed polarity and gives a paleolatitude of 34°±8°. The Alitak Bay locality has only reversed polarity and gives a paleolatitude 49°±8° [Plumley et al., 1983, Coe et al., 1985]. The paleolatitude discrepancies could be the result of problems in measuring structural attitudes, initial dip error or incomplete secular variation averaging. Coe et al. [1985] have considered these possibilities and believe that a paleolatitude of 40°±6° (25°±7° displacement) given by the combined data is more probable than different paleolatitudes for the two localities. However, the possibility of a significant age difference between the sample localities cannot be ruled out.

The age of magnetization for the Ghost Rocks is considered to be Paleocene (62-66 Ma) based on regional age constraints of the formation rather than age constraints established at the individual outcrops sampled for paleomagnetism. While there are Paleocene fossils in the Ghost Rocks

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TABLE 4. Summary of Poles with Stability Tests

Location,	Reference	Age	STRAT				VGP				Northward Displacement	Rotation
			D	I	K	α_5	Longitude	Latitude	K	α_{95}		
Talkeetna Mts	1	47Ma	301	84	36.7	4.6	188	66	11.4	8.4	-13±8	-50±40
Talkeetna Mts	2	54	137	-82	16.0	7.3	180	70	6	12.2	-8±10	-37±44
Cantwell	3	(58)	(143)	(-85)	(43.1)	(5.0)	(195)	(70)		(10.0)	(-9±8)	(-129±?)
Cantwell	4	60	148	-79	24.6	7.9	161	76		9.8	+2±8	-32±25
Alitak	5	62-66	82	65	25.8	7.4	269	43	13.0	10.6	16±9	86±14
Kiluida	5	62-66	320	53	29.4	8.6	93	55	19.8	10.5	31±9	-34±12
Lake Clark	6	66	299	74	32.8	4.7	147	61	12.0	7.9	10±8	-54±22
Bristol Bay	6, 7	68	307	76	31.4	3.0	147	65	11.7	5.0	9±7	-43±23
Resurrection Pen.	6, 8	70					170	32		13.0	24±12	-40±18
Resurrection Pen.	9	60									13±6	
Carmacks Volcanics	10	70	167	-71	53	4.8	109	82	21	7.8	13±8	10±21
MacColl Ridge	11	74	226	51	20.3	9.1	180	12	14.8	10.7	42±9	-107±22
St. Matthew Is.	12	75	26	74	65		240	76	31	9	13±10	22±34
Koyukuk	13	90	349	77	53	6	132	85	17	11	17±9	70±35
Brooks Range	14	100					156	76		6.3	9±6	18±46
Koyukuk	15	140									16-24±6	

See Table 2 footnotes. Absence of data indicates that the original reference did not report the values. Questionable confidence limits for the amount of rotation are difficult to assess due to the steepness of the recorded geomagnetic field and are at least as large as the amount of rotation indicated. Ages have several m.y. uncertainty. Approximate ages have larger uncertainties due to imprecise paleontological dating and geological relationships. Sources of data as follows: 1, this study; 2, Hillhouse et al. [1985]; 3, Hillhouse and Grommé [1982]; 4, Panuska and Macicak [1986]; 5, Plumley et al. [1983]; 6, [Coe et al. [1985]; 7, Globerman [1985] 8, Hillhouse and Grommé [1977]; 9, Bol and Coe [1987]; 10, Marquis and Globerman [1988]; 11,Panuska [1984, 1985]; 12, Wittbrodt [1985]; 13, Harris et al. [1987]; 14, Witte et al. [1987]; 15, Hillhouse and Grommé [1985].

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Formation, there is also an Upper Cretaceous fossil locality [Moore et al., 1983]. If the ages of magnetization of the two localities are in fact different, it is possible that the different paleolatitudes calculated are in fact real. In this scenario the different paleolatitudes would provide a measure of displacement versus time.

The summary data in Figure 9 show some interesting trends. It seems clear that the inboard terranes were located approximately 10° south of the their present day position during the Cretaceous. These terranes appear to have been displaced to the north some time during the latest Cretaceous and earliest Tertiary. During this time interval, the terranes of the SAS were located at least as far south as the inboard terranes. Given the uncertainties in age, secular variation averaging and inclination error, the SAS data are difficult to interpret with any confidence.

If the lowest displacement values, within the error limits for the SAS are correct, there may be no significant paleolatitude difference between the SAS and the inboard terranes. This scenario has the advantage of being able to account for most or all of the northward displacement by strike-slip along the Tintina [e.g., Gabrielse, 1985] and Denali fault systems. The MacColl Ridge sandstone data would then have to be discounted on the basis of an inferred inclination error. Alternatively, if the larger displacement values for the Resurrection and Ghost Rocks data are correct, then there would be some evidence for the rapid northward displacement suggested by Panuska [1985]. Estimates of the rapid northward displacement rates are roughly similar to the Kula plate velocity estimates [Engebretson et al., 1985; Rea and Duncan, 1986]; however, geologic evidence for such large displacements has not been identified. Given the uncertainties, paleomagnetic interpretation will have to be deferred until more data with tight age control can be acquired.

A few additional implications of the Talkeetna data deserve some mention. The paleomagnetic pole for the southern Talkeetna Mountains is displaced from the 50 Ma North American reference pole in a counterclockwise sense. This declination anomaly yields an estimate of 50°±40° counterclockwise rotation. This is in agreement with other paleomagnetic studies which show 30°-50° counterclockwise rotation for southern and central Alaska (south of the Kaltag fault). Moreover, these are the youngest rocks to record significant counterclockwise rotation. The existence of anomalous declinations appears to be well established regional phenomenon, although the mechanism of rotation remains a topic of debate [e.g., Hillhouse et al., 1985; Coe and Thrupp, 1987; Panuska, 1987; Coe et al., 1989].

In addition to anomalous declination, the two Talkeetna data sets show anomalously high paleolatitudes. The northern Talkeetna data suggest a paleolatitude 8° higher than expected, however, this value is not statistically significant. The southern Talkeetna data yield a paleolatitude 13° higher than expected which appears to be only marginally significant. While the displacement curve (Figure 9) suggests northward motion into early Tertiary time with an "overshoot," it remains to be seen if this effect is real or an artifact of high paleolatitude, incomplete secular variation averaging and imprecise age control.

Appendix

The paleolatitude error bars are simply the 95% confidence limits based on the scatter of the paleomagnetic data. Error limits for the age control (in Figure 9) employ diverse types of data and geological arguments. This requires special evaluation on a case by case basis.

The MacColl Ridge datum is assigned an age of 74 Ma±4 Ma. The MacColl Ridge Formation was considered to be Campanian or Maastrichtian because it conformably overlies the Campanian Chititu Formation [Jones and MacKevett, 1969]; however, no age diagnostic fossils were found in the MacColl Ridge. Panuska [1984, 1985] considered the age to be early Maastrichtian on the basis of mixed polarity evidence, since the lower Campanian is reversed polarity and the upper Campanian is of normal polarity [Lowrie and Alvarez, 1981]. The error limits assigned to the age reflect the uncertainty of Maastrichtian-Campanian stage boundary reported in the DNAG Geologic Time Scale.

The Resurrection Peninsula data come from pillow lavas and dikes within the Valdez(?) Group [Hillhouse and Grommé, 1977]. The Valdez Group is assigned a Maastrichtian age based on fossil evidence [Jones and Clark, 1973]. The median age of the Maastrichtian, 70 Ma, may be assigned to the paleolatitude estimate with the error limits being the duration of the Maastrichtian. The plotted error limits are queried because of uncertain correlation.

The age of the Alitak Bay and Kiliuda Bay paleolatitude is approximately 62-66 Ma. These localities are part of the Ghost Rocks Formation which contain Paleocene fossils and are intruded by 62 Ma plutonic rocks [Plumley et al., 1983; Moore et al., 1983].

Volcanic rocks from the Lake Clark area have only one K-Ar age determination, with high analytical uncertainty (66±14 Ma), constraining the age of magnetization [Coe et al., 1985]. Thrupp and Coe [1986] consider an early Paleocene age to be corroborated by other K-Ar analyses from the area.

The rocks sampled by Hillhouse et al. (1985) have two K-Ar ages of 53.9±1.6 Ma and 51.3±2.5 Ma. Associated rocks from the area give several K-Ar ages ranging from 50.4 Ma to 58.6 Ma [Hillhouse et al., 1985]. This age range is adopted as the error limits for the age of magnetization.

Acknowledgments. We thank Robin Cottrell for technical assistance with the K and Ar analyses, Paul Wittbrodt for field assistance and Malcolm Johnson for technical help with X ray and SEM work. BP would like to thank Jerry Bock and Helaine Pleet for their hospitality during the lab work. Special thanks are owed Mike Gardner and Steve Bergman of ARCO Exploration and Technology for logistic support. Financial support for part of the project was supplied by B.P. Alaska.

References

Bol, A.J., and R.S. Coe, Paleomagnetism of pillow basalt and sheeted dikes, Resurrection Peninsula, Alaska: Confirmation of large northward displacement in S. Alaska since

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Early Tertiary, Geol. Soc. Am. Abstr. Programs, 19(7), 594, 1987.
Coe, A. S., B. R. Globerman, and G. A. Thrupp, Rotation of central and southern Alaska in the early Tertiary: oroclinal bending by megakinking? in Paleomagnetic Rotations and Continental Deformation, edited by C. Kissel and C. Laj, Kluwer Academic Publishers, 327-342, 1989.
Coe, R.S., and G.A. Thrupp, Tectonic implications of rotated paleomagnetic declinations in west-central Alaska, Geol. Soc. Am. Abstr. Programs, 19(6), 367, 1987.
Coe, R.S., B.R. Globerman, P.W. Plumley, and G.A. Thrupp, Paleomagnetic results from Alaska and their tectonic implications, in Tectonostratigraphic Terranes of the Circum-Pacific Region, Earth Sci. Ser., Vol. 1, edited by D. G. Howell, pp. 85-108, Circum-Pacific Council for Energy and Mineral Resources, Houston, Tex., 1985.
Csejtey, B., D. P. Cox, R. C. Evarts, G. D. Stricker, H. L. Foster, The Cenozoic Denali Fault System and the Cretaceous accretionary development of southern Alaska, J. Geophys. Res., 87, 3,741-3,754, 1982.
Detterman, P.C., G. Plafker, R.G. Tysdal, and T. Hudson, Geology and surface features along part of the Talkeetna segment of the Castle Mountain-Caribou fault system, scale 1:63, 360, U.S. Geol. Surv. Misc. Field Stud. Map, MF-738,1976.
Diehl, J.F., M.E. Beck, S. Beske-Diehl, D. Jacobson, and B.C. Hearn, Paleomagnetism of the Late Cretaceous-Early Tertiary central Montana alkalic province, J. Geophys. Res., 88, 10,594-10,609, 1983.
Engebretson, D.C., A. Cox, and R. G. Gordon, Relative motions between oceanic and continental plates in the Pacific Basin, Spec. Pap. Geol. Soc. Am., 206, 1985.
Gabrielse, H., Major dextral transcurrent displacements along the Northern Rocky Mountain Trench and related lineaments in north-central British Columbia, Geol. Soc. Am. Bull., 96, 1-14, 1985.
Globerman, B.R., A paleomagnetic and geochemical study of Upper Cretaceous to Lower Tertiary rocks from the Bristol Bay Region, southwestern Alaska, Ph.D. dissertation, 292 pp., Univ. of Calif., Santa Cruz, 1985.
Grantz, A., Geologic map of Talkeetna Mountains (A-2) quadrangle, Alaska and the contiguous area to the north and northwest, scale 1:48,000, U.S. Geol. Surv. Misc. Geol. Invest. Map, I-313, 1960a.
Grantz, A., Geologic map of Talkeetna mountains (A-1) Quadrangle, and the south third of Talkeetna Mountains (B-1) Quadrangle, Alaska, scale 1:48,000, U.S. Geol. Surv. Misc. Geol. Invest. Map, I-314, 1960b.
Grantz, A., Generalized geologic map of the Nelchina area, Alaska, showing igneous rocks and Langer faults, scale 1:96,000, U.S. Geol. Surv. Misc. Geol. Invest. Map, I-312, 1960c.
Grantz, A., Geologic map and cross sections of the Anchorage (D-2) Quadrangle and northeastern most part of the Anchorage (D-3) Quadrangle, Alaska, scale 1:48,000, U.S. Geol. Surv. Misc. Geol. Invest. Map, I-342, 1961.
Haeussler, P. J., and R. S. Coe, New paleomagnetic results from the Hound Island volcanics and implications for the Alexander Terrane, Geol. Soc. Am. Abstr. Programs, 19(6), 385, 1987.
Harris, R.A., D.B. Stone, and D.L. Turner, Tectonic implications of paleomagnetic and geochronologic data from the Yukon-Koyukuk Province, Alaska, Geol. Soc. Am. Bull., 99, 362-378, 1987.
Hillhouse, J.W., Paleomagnetism of the Triassic Nikolai Greenstone, McCarthy Quadrangle Alaska, Can. J. Earth Sci., 14, 2578-2592, 1977.
Hillhouse, J.W., and C.S. Grommé, Paleomagnetic poles from sheeted dikes and pillow basalt of the Valdez and Orca groups, southern Alaska, Eos Trans. AGU, 58, 1127, 1977.
Hillhouse, J.W., and C.S. Grommé, Limits to northward drift of the Paleocene Cantwell Formation, central Alaska, Geology, 10, 552-556, 1982.
Hillhouse, J.W., and C.S. Grommé, Early Cretaceous paleolatitude of the Yukon-Koyukuk Basin, Eos Trans. AGU, 66, 1103, 1985.
Hillhouse, J.W., C.S. Grommé, and B. Csejtey, Tectonic implications of paleomagnetic poles from Lower Tertiary volcanic rocks, south central Alaska, J. Geophys. Res., 90, 12,523-12,535, 1985.
Jones, D.L., and S.H.B. Clark, Upper Cretaceous (Maastrichtian) fossils from the Kenai-Chugach Mountains, Kodiak and Shumagin islands, southern Alaska, U. S. Geol. Surv., J. Res., 1(2), 125-136, 1973.
Jones, D.L., and E.M. MacKevett, Summary of Cretaceous stratigraphy in part of the McCarthy Quadrangle, Alaska, U. S. Geol. Surv. Bull., 1274-K, K1-K19, 1969.
Jones, D.L., N.J. Silberling, and J. Hillhouse, Wrangellia--A displaced terrane in northwestern North America, Can. J. Earth Sci., 14, 2565-2577, 1977.
Lowrie, M., and W. Alvarez, One hundred million years of geomagnetic polarity history, Geology, 9, 392-397, 1981.
Magoon, L.B., W.L. Adkison, and R.M. Egbert, Map showing geology wildcat wells, Tertiary plant fossil localities, K-Ar age dates, and petroleum operations, Cook Inlet Area, Alaska, scale 1:250,000, U.S. Geol. Surv. Misc. Invest. Ser., 1976.
Marquis, G., and B. R. Globerman, Northward motion of the Whitehorse Trough: Paleomagnetic evidence from the Upper Cretaceous Carmacks Group, Can. J. Earth Sci., 25, 2005-2016, 1988.
McFadden, P.L., and D.L. Jones, The fold test in paleomagnetism, Geophys. J. R. Astron. Soc., 67, 53-58, 1981.
Moore, J.C., T. Byrne, P.W. Plumley, M. Reid, H. Gibbons, and R.S. Coe, Paleogene evolution of the Kodiak Islands, Alaska: Consequences of ridge-trench interaction in a more southerly latitude, Tectonics, 2, 265-293, 1983.
O'Reilly, W., Rock and Mineral Magnetism, 220 pp., Blackie, Glasgow, 1984.
Panuska, B.C., Paleomagnetism of the Wrangellia and Alexander terranes and the tectonic history of southern Alaska, Ph.D. thesis, 197 pp., Univ. of Alaska, Fairbanks, 1984.
Panuska, B.C., Paleomagnetic evidence for a post-

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Cretaceous accretation of Wrangellia, Geology, 13, 880-883, 1985.
Panuska, B.C., Is there an orocline in south central Alaska?, Geol. Soc. Am. Abstr. Programs, 19(6), 438, 1987.
Panuska, B.C., and J. Decker, Reconnaissance paleomagnetic study of the Eocene Admiralty Island Volcanics, southeast Alaska: Evidence for pre-late Eocene accretaion, Geol. Soc. Am. Abstr. Programs, 17(7), 684, 1985.
Panuska, B.C., and M. Macicak, A revised paleolatitude for the Paleocene Teklanika Volcanics, central Alaska Range, Alaska, Eos Trans. AGU, 67, 921, 1986.
Panuska, B.C., and D.B. Stone, Latitudinal motions of the Wrangellia and Alexander terranes and the southern Alaska superterrane, in Tectonostratigraphic Terranes of the Circum-Pacific Region, Earth Sci. Ser., edited by D.G. Howell, pp. 109-120, Vol. 1, Circum-Pacific Council for Energy and Mineral Resources, Houston, Tex., 1985.
Plumley, P.W., R.S. Coe, and T. Byrne, Paleomagnetism of the Paleocene Ghost Rocks Formation, Prince William Terrane, Alaska, Tectonics, 2, 295-314, 1983.
Rea, D.K., and R.A. Duncan, North Pacific plate convergence: A quantitative record of the past 140 million years, Geology, 14, 373-376, 1986.
Shive, P.N., Suggestions for the use of SI units in magnetism, Eos Trans. AGU, 67, 25, 1986.
Silberman, M.L., and A. Grantz, Paleogene volcanic rocks of the Matanuska Valley area and the displacement history of the Castle Mountain fault, The United States Geological Survey in Alaska: Accomplishments During 1981, edited by W. L. Conrad and R. L. Elliott, U. S. Geol. Surv. Circ., 868, 82-86, 1984.
Thrupp, G.A., and R.S. Coe, Early Tertiary paleomagnetic evidence and the displacement of southern Alaska, Geology, 14, 213-217, 1986.
Wittbrodt, P., Paleomagnetism and petrology of St. Matthew Island, Bering Sea, M.S. thesis, 210 pp., Univ. of Alaska, Fairbanks, 1985.
Witte, W.K., D.B. Stone, and C.G. Mull, Paleomagnetism, paleobotany and paleogeography of the Cretaceous, Arctic Slope, Alaska, in Alaskan North Slope Geology, edited by I. Tailleur and P. Weiner, pp. 771-780, Society of Economic Paleontologists and Mineralogists/Alaska Geological Society, Anchorage, 1987.

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(Received February 27, 1989;
revised October 27, 1989;
accepted November 1, 1989.)

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